CSCE 5585 LAB 4

Advanced Wireshark Lab on NAT and Protocol Analysis

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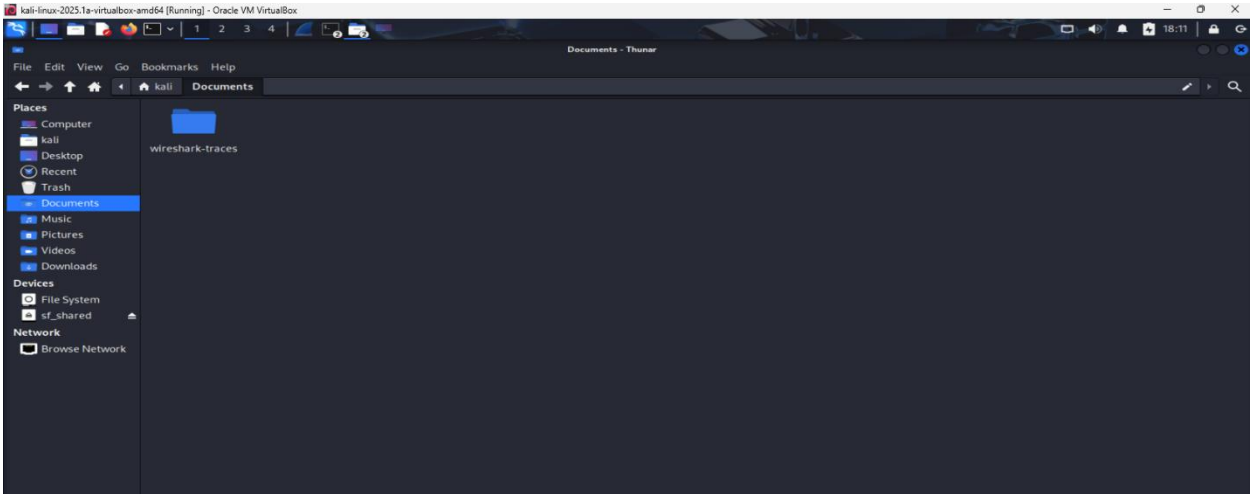
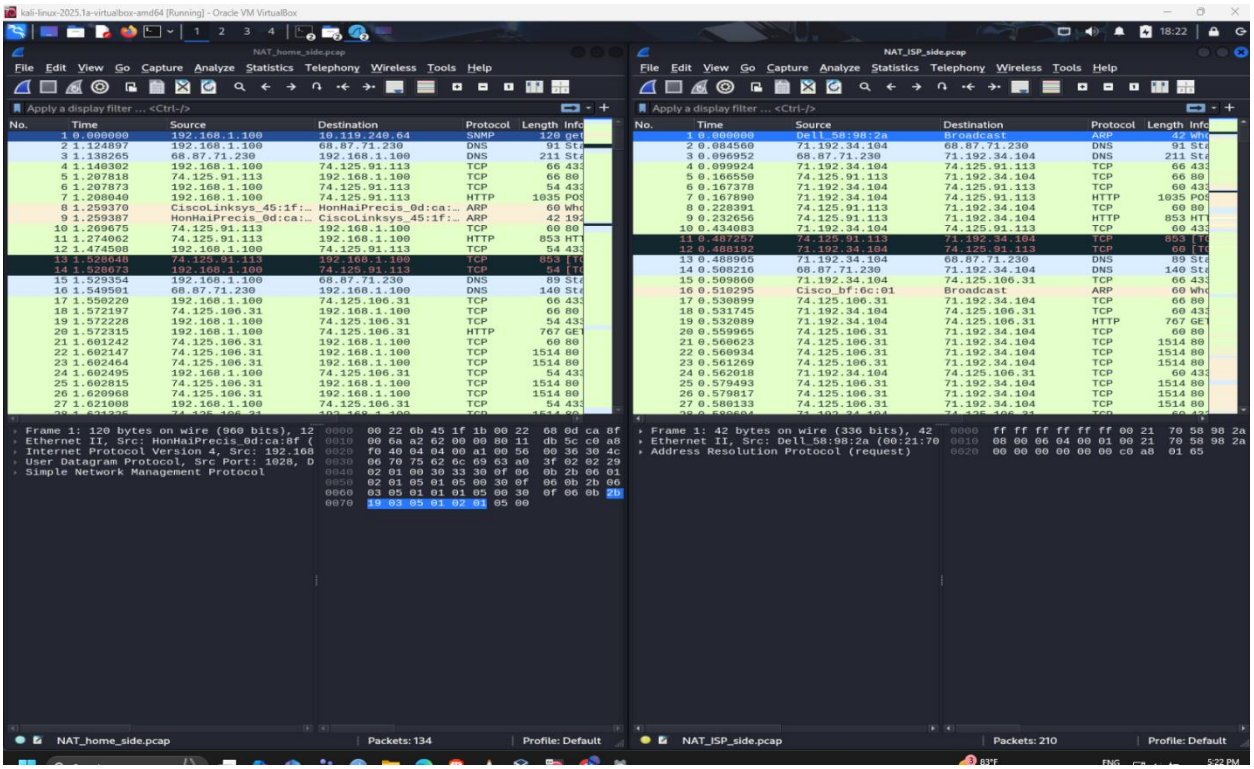
Part1: NAT Measurement Scenarios

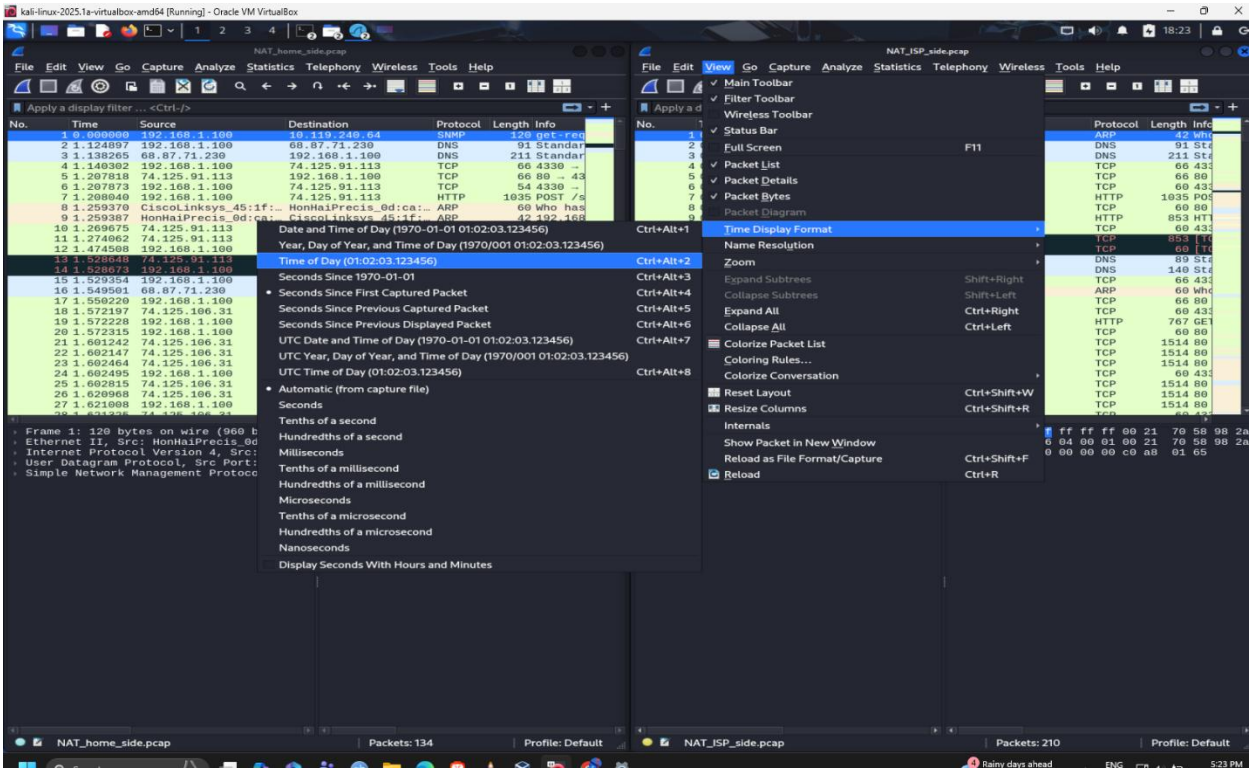
Downloaded the wireshark logs from canvas



Opening logs both side by side NAT_home_side.pcap and NAT_ISP_side.pcap

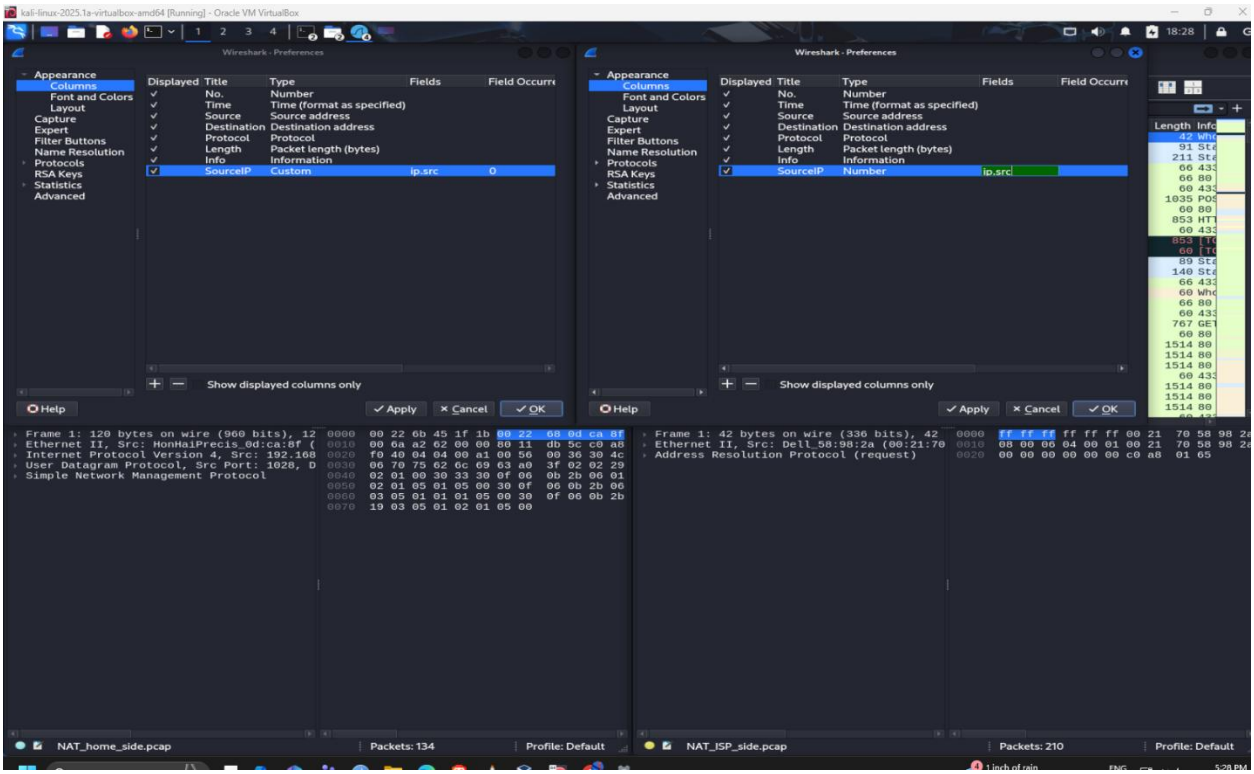


For correct format for comparing packet timing later.



Add required columns

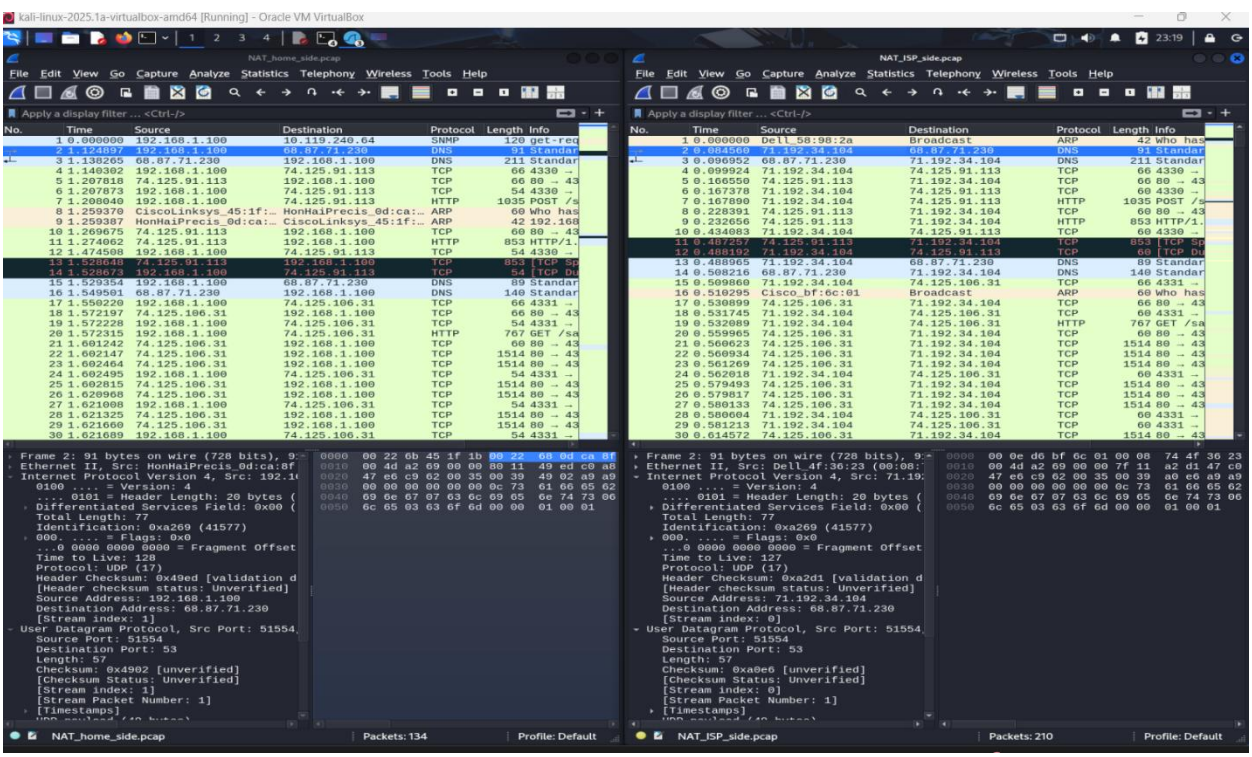
Title: Source IP **custom column** ip.src



Part 2: TCP Strem Analysis

IP Comparison and NAT Changes:

From the screenshot below



Details of frame 2 from NAT_home_side.pcap

2 1.124897 192.168.1.100 68.87.71.230 DNS 91 Standard query 0xa9a9 A safebrowsing.clients.google.com 192.168.1.100

Details from Frame 2 packet capture from NAT_ISP_side.pcap

2 0.084560 71.192.34.104 68.87.71.230 DNS 91 Standard query 0xa9a9 A safebrowsing.clients.google.com 71.192.34.104

We can clearly see

192.168.1.100 → 68.87.71.230 DNS Standard query 0xa9a9 **Asafebrowsing.clients.google.com**

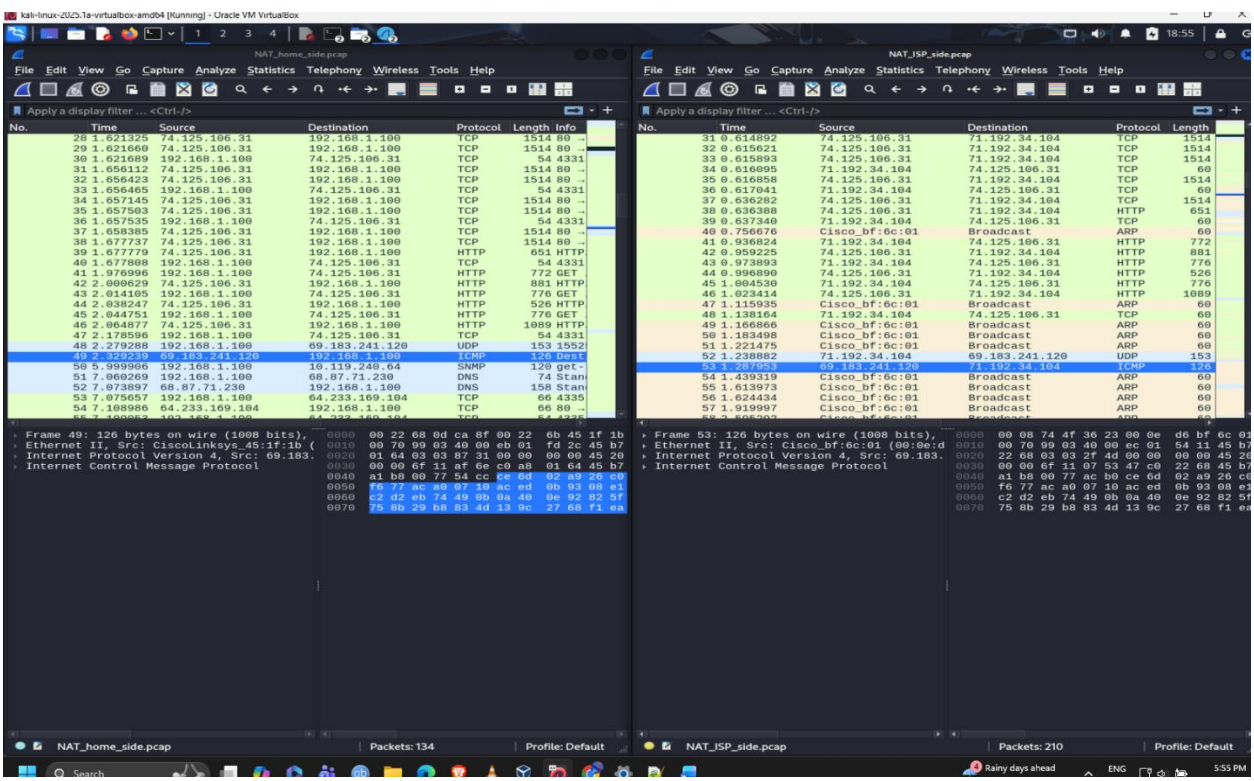
71.192.34.104 → 68.87.71.230 DNS Standard query 0xa9a9 **Asafebrowsing.clients.google.com**

A DNS query is received by the internal host 192.168.1.100. At the ISP comp the packet is received with a source IP 71.192.34.104. with transaction ID and the destination of 68.87.71.230 are equals indicating that the NAT equipment just substituted the source IP.

Another example of this we can see

UDP + ICMP example (port unreachable)

From screenshot below



Home side:

Frame 48 2.279288 192.168.1.100 69.183.241.120 UDP 153 15525 → 41400 Len=111 192.168.1.100

192.168.1.100:15525 → 69.183.241.120:41400 UDP

Frame 49 2.329239 69.183.241.120 192.168.1.100 ICMP 126 Destination unreachable (Port unreachable) 69.183.241.120,192.168.1.100

69.183.241.120 → 192.168.1.100 ICMP Destination unreachable (Port unreachable)

ISP side:

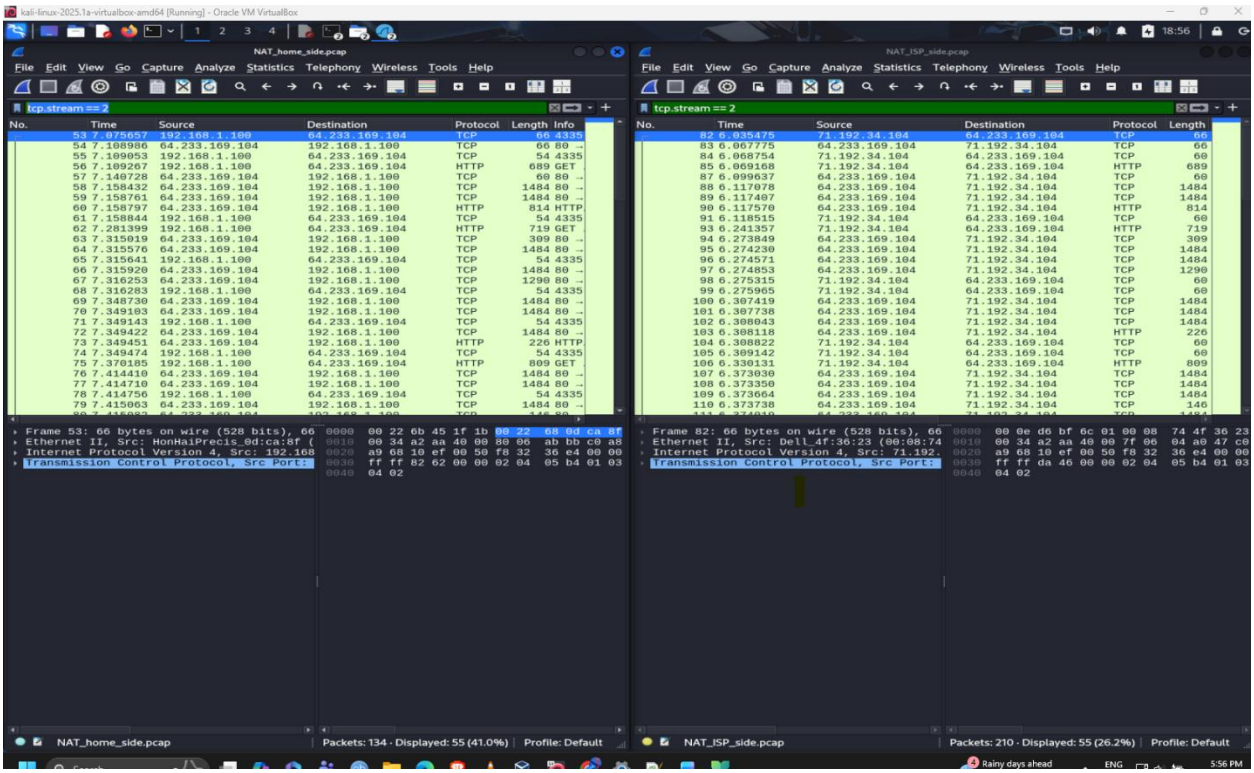
Frame 52: 52 1.238882 71.192.34.104 69.183.241.120 UDP 153 15525 → 41400 Len=111 71.192.34.104

71.192.34.104:15525 → 69.183.241.120:41400 UDP

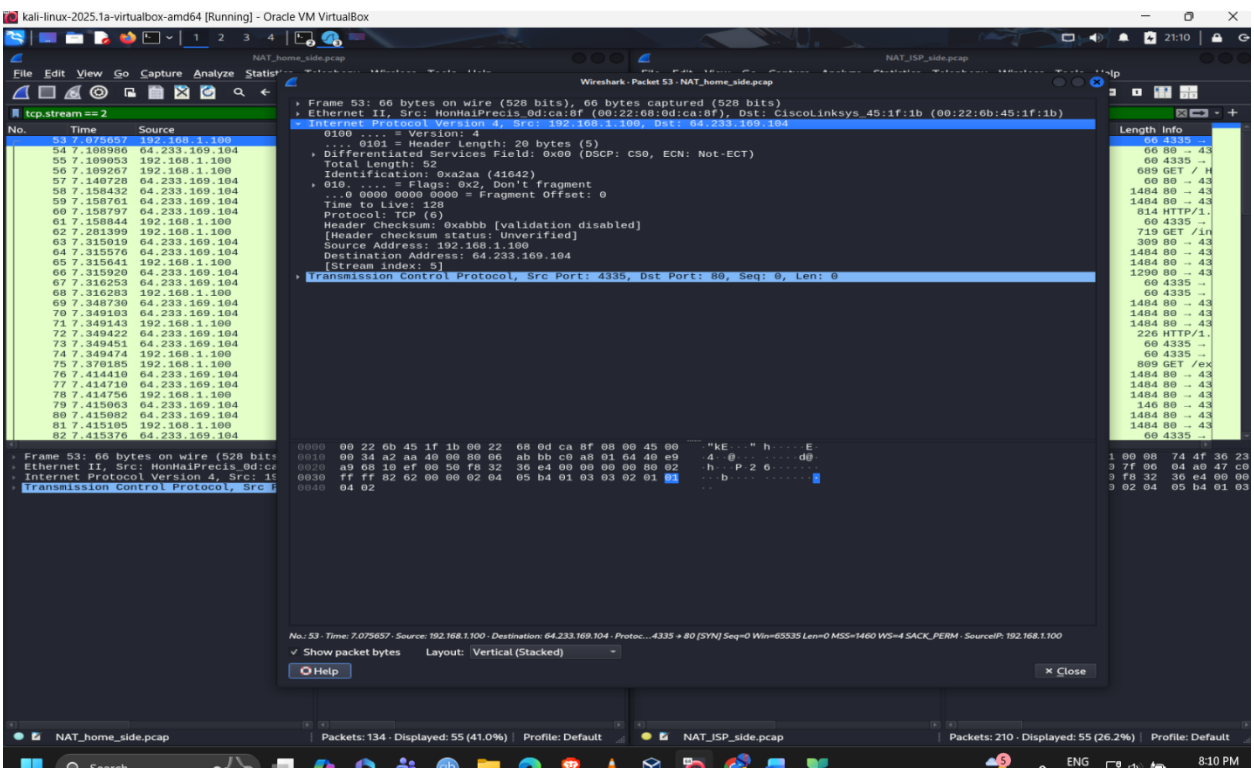
Frame 53: 1.287953 69.183.241.120 71.192.34.104 ICMP 126 Destination unreachable (Port unreachable) 69.183.241.120,71.192.34.104

69.183.241.120 → 71.192.34.104 ICMP Destination unreachable

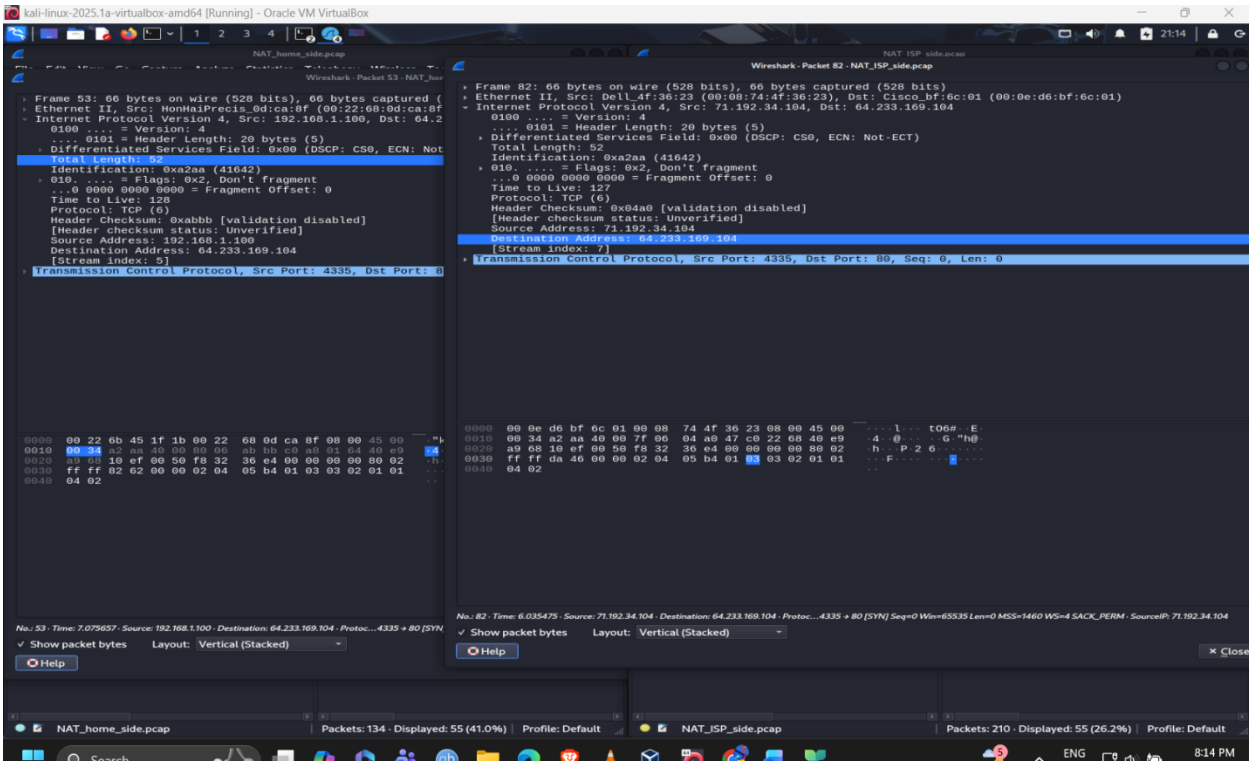
Here we can see that even error messages (ICMP) are sent back to the public IP, then the NAT translates them and delivers back them to the internal host.
From screenshot below



Homeside.pcap



Frame 53 7.075657 192.168.1.100 64.233.169.104 TCP 66 4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM 192.168.1.100



ISP-side.pcap

Frame 82 6.035475 71.192.34.104 64.233.169.104 TCP 66 4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM 71.192.34.104

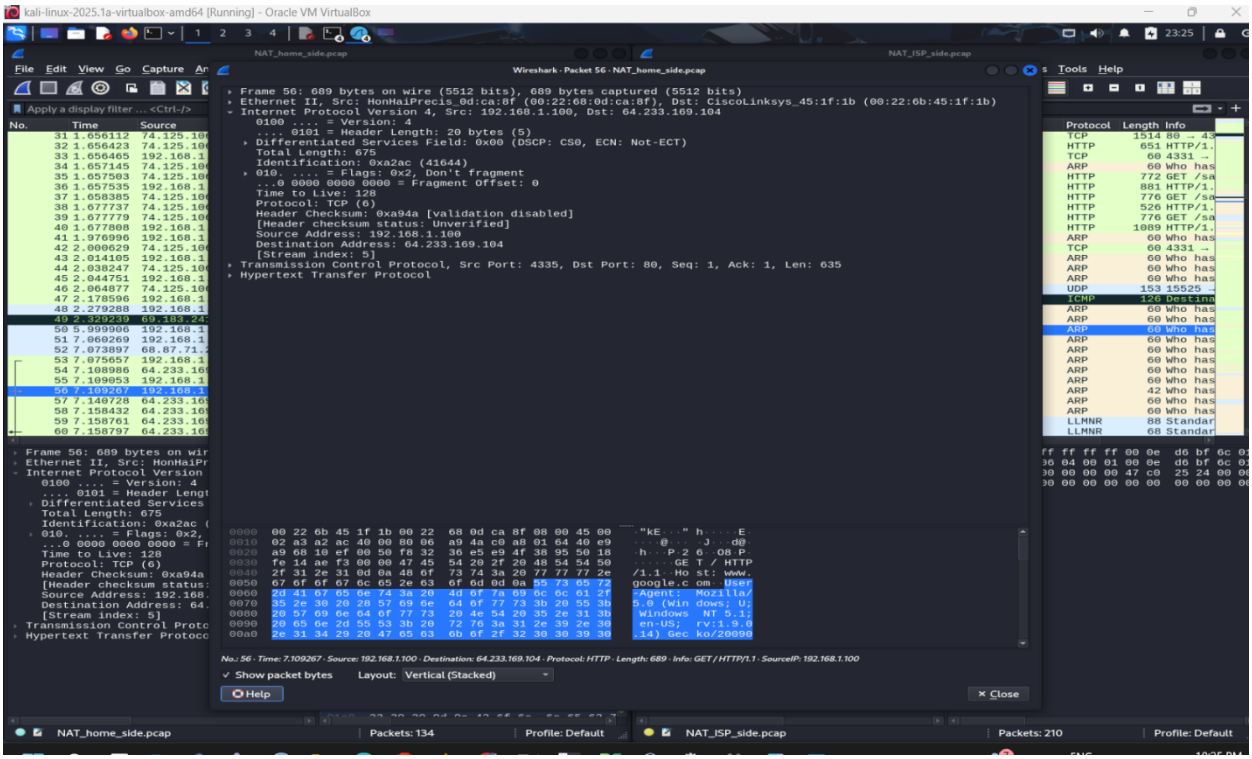
Here, after careful observation we can see that

NAT changed only the client IP address

192.168.1.100 → 71.192.34.104

The client port (4335) and the server IP/port stay the same.

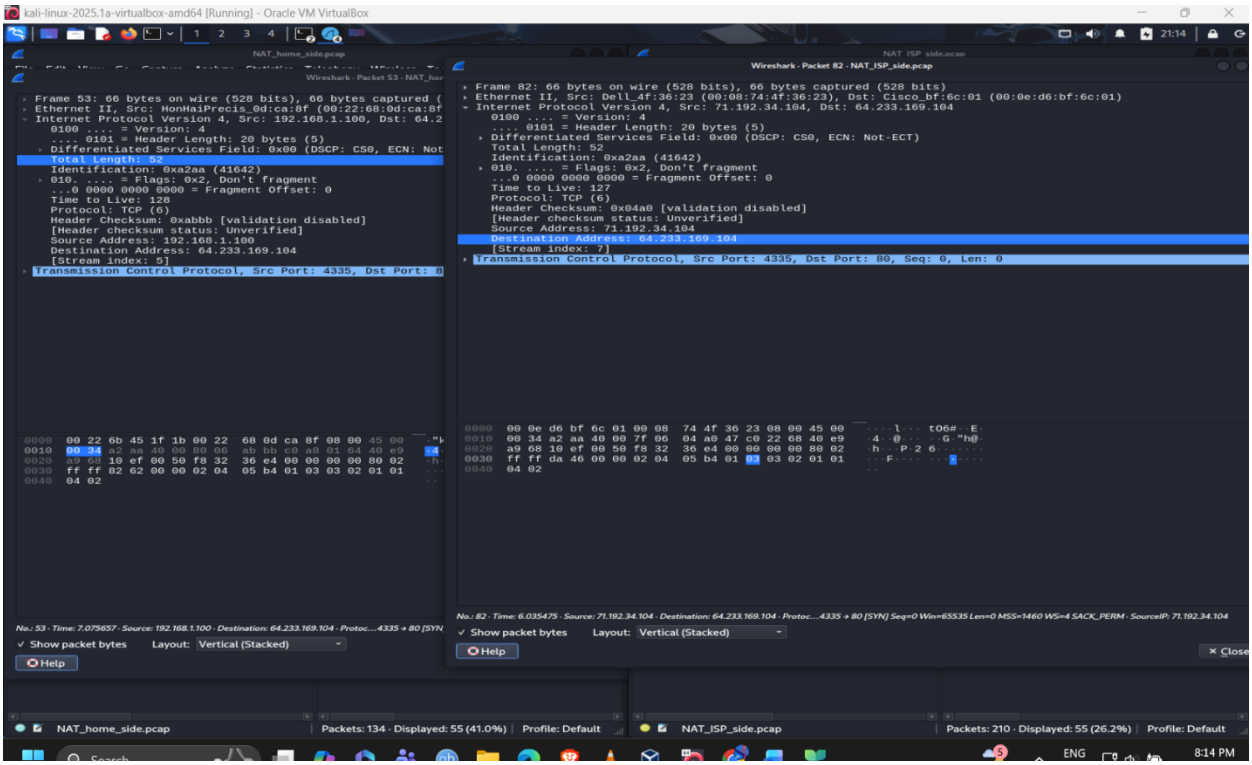
In other example HTTP GET /



Home: frame 56

56 7.109267 192.168.1.100 64.233.169.104 HTTP 689 GET / HTTP/1.1 192.168.1.100

192.168.1.100:4335 → 64.233.169.104:80 HTTP GET /



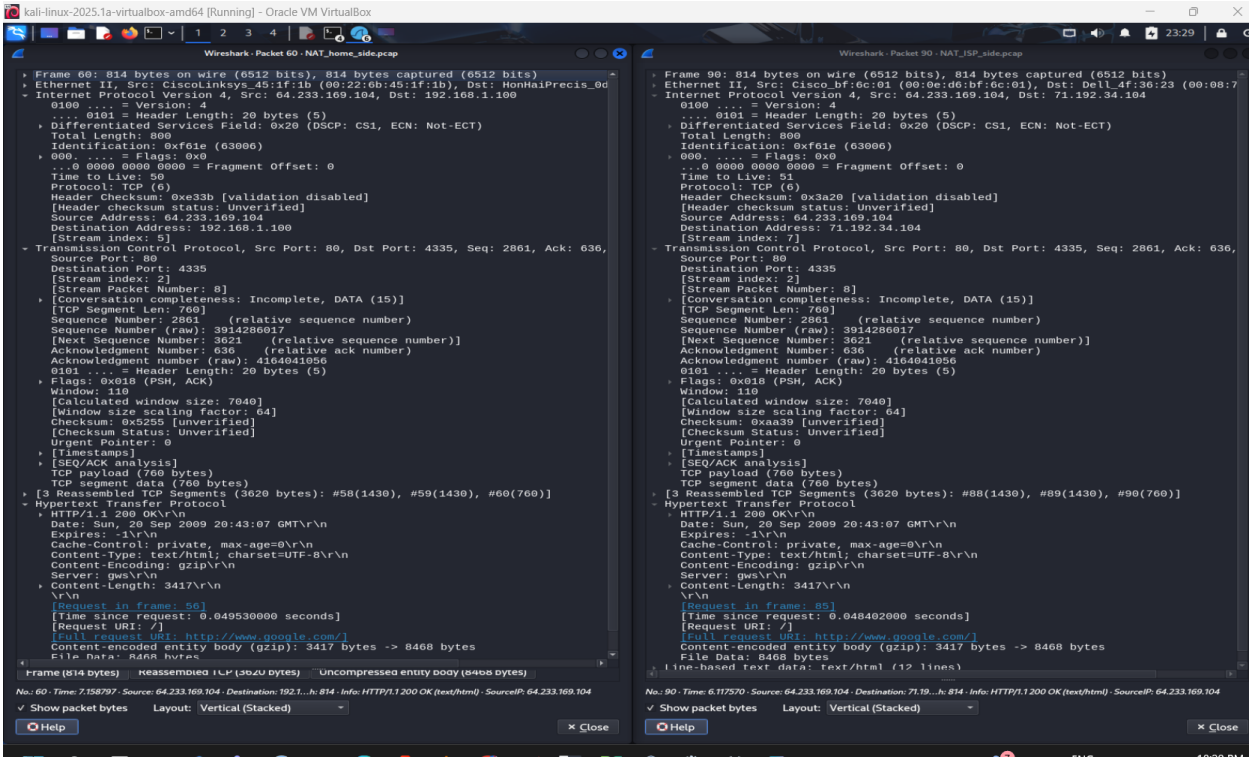
ISP: frame 85

85 6.069168 71.192.34.104 64.233.169.104 HTTP 689 GET / HTTP/1.1 71.192.34.104

71.192.34.104:4335 → 64.233.169.104:80 HTTP GET /

from other first HTTP 200 OK

Home: frame 60



60 7.158797 64.233.169.104 192.168.1.100 HTTP 814 HTTP/1.1 200 OK (text/html) 64.233.169.104

64.233.169.104 → 192.168.1.100 HTTP/1.1 200 OK

ISP: frame 90

90 6.117570 64.233.169.104 71.192.34.104 HTTP 814 HTTP/1.1 200 OK (text/html) 64.233.169.104

64.233.169.104 → 71.192.34.104 HTTP/1.1 200 OK

Q. TCP Header analysis

the private/internal IP and port?

→ 192.168.1.100:4335

the public/translated IP and port?

→ 71.192.34.104:4335

The fields stay the same across the NAT?

→ Server IP (64.233.169.104), server port (80), client port (4335), TCP seq/ack numbers, payload (HTTP GET, HTTP 200), etc.

Field changes?
→ The client IP address (192.168.1.100 ↔ 71.192.34.104)

All TCP header fields like ports, flags, sequence number, window and options are the same between the home-side and ISP-side first SYN packet, the mode of difference is the value of TCP checksum that will be recalculated by NAT due to the coverage of the IP pseudo header and change in source IP.

IP Header Analysis

Still on the first SYN packet in each trace:

Internet Protocol Version 4.

Fields we are looking at

Source IP:

Home: 192.168.1.100

ISP: 71.192.34.104 changed by NAT

Destination IP: 64.233.169.104 (same)

Protocol: TCP (6) (same)

Total Length, ID: usually same or very close.

TTL: often decreased by 1 at the ISP side (router hop).

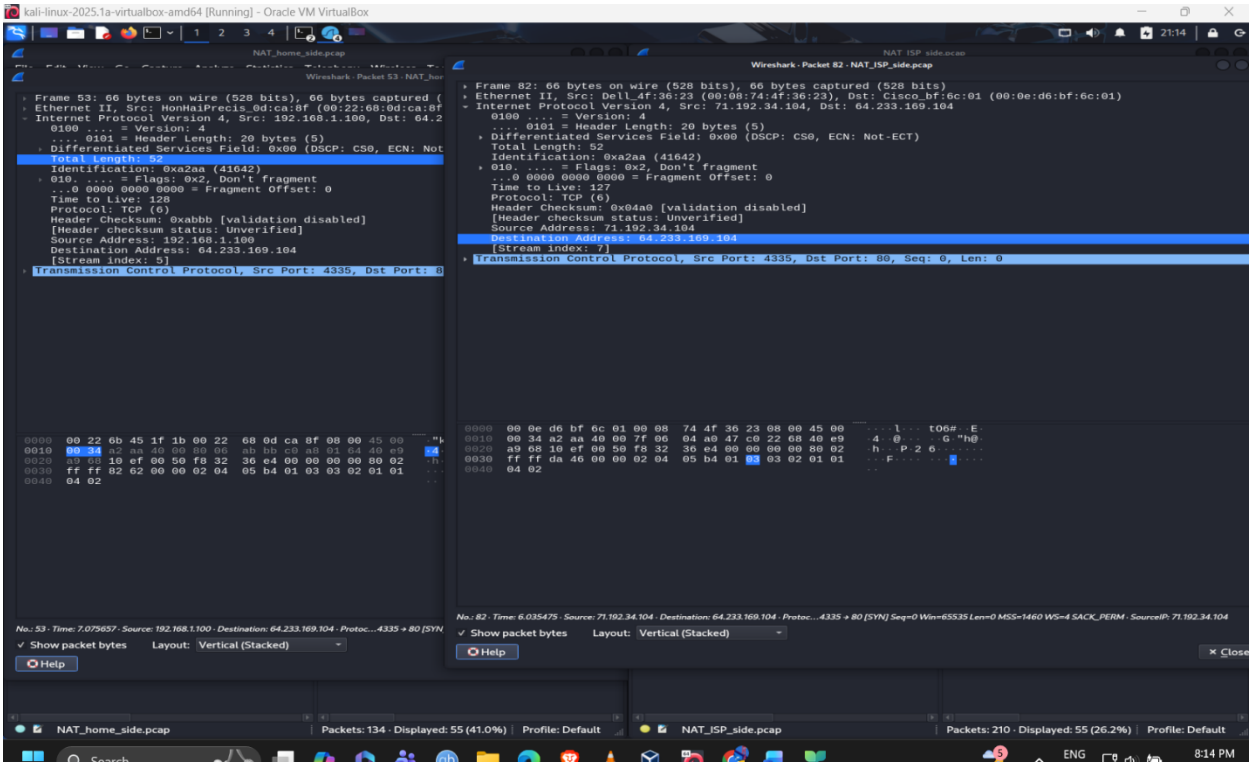
Header checksum: different (recomputed by router/NAT).

Source IP address, changed from private 192.168.1.100 to public 71.192.34.104 so the packet is routable on the Internet.

TTLdecreased by 1 because each router hop decrements TTL.

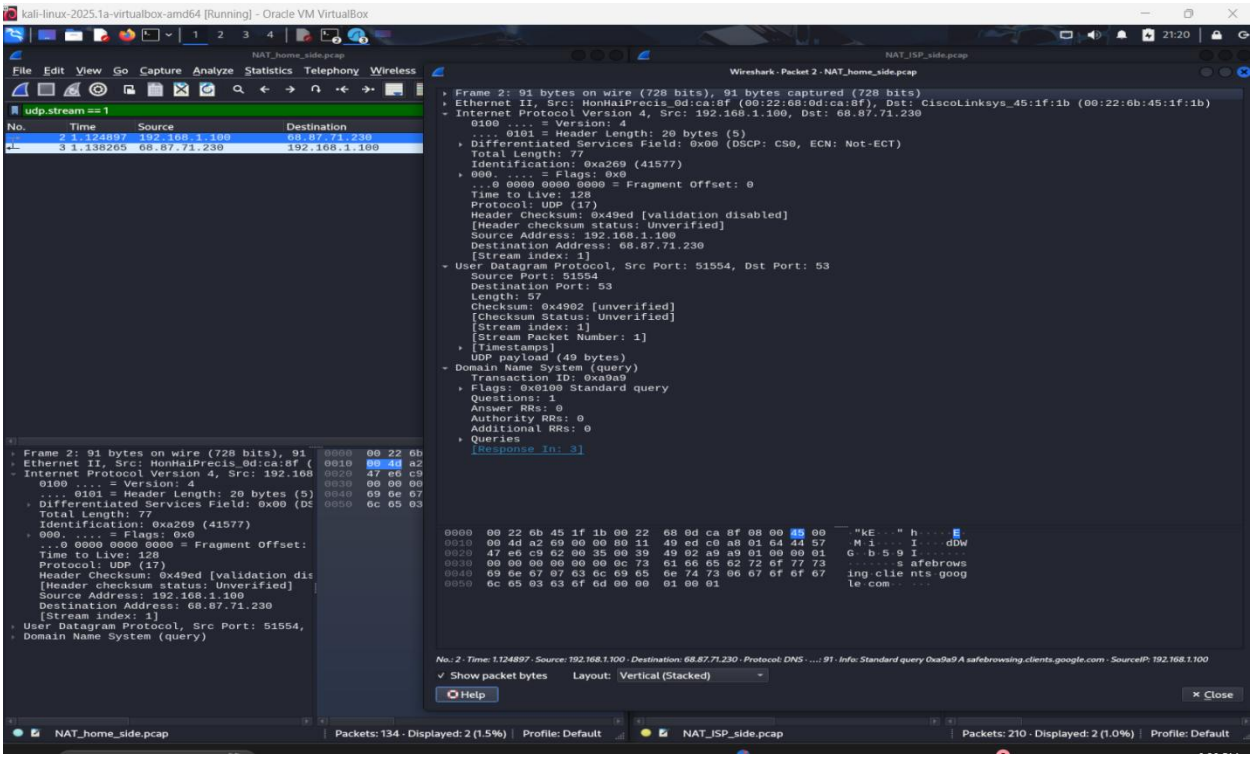
IP header checksum is recalculated because the header source IP and TTL changed, when NAT changes the source IP, it **must recompute** the checksum so the packet remains valid.

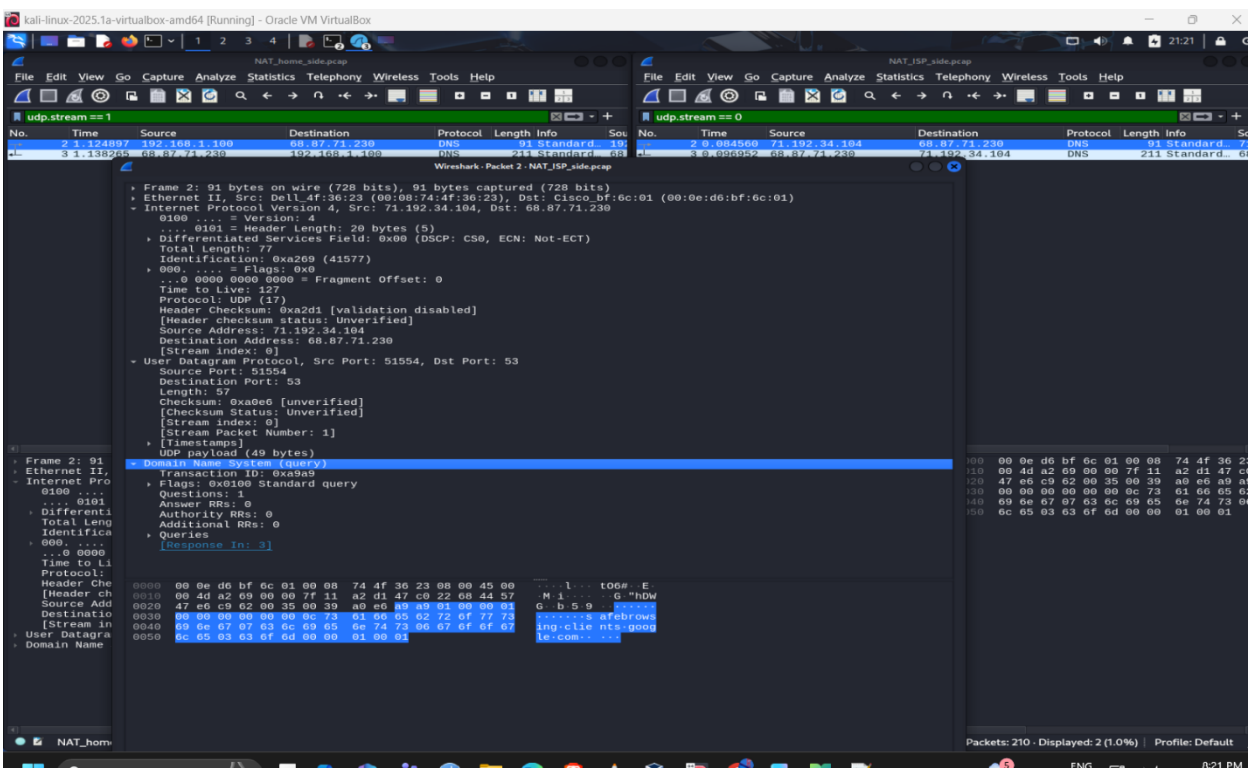
Everything else dest IP, protocol, total length is the same.


kali-linux-2025.1a-virtualbox-amd64 [Running] - Oracle VM VirtualBox
NAT_home_side.pcap
Wireshark · Packet 53 · NAT_hor
Frame 53: 66 bytes on wire (528 bits), 66 bytes captured (
Ethernet II, Src: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f
Internet Protocol Version 4, Src: 192.168.1.100, Dst: 64.2
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not
Total Length: 52
Identification: 0xa2aa (41642)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0xabbb [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.100
Destination Address: 64.233.169.104
[Stream index: 5]
Transmission Control Protocol, Src Port: 4335, Dst Port: 8
0000 00 22 6b 45 1f 1b 00 22 68 0d ca 8f 08 00 45 00
0010 00 34 a2 aa 40 00 80 06 ab bb c0 a8 01 64 40 e9
0020 a9 68 10 ef 00 50 f8 32 36 e4 00 00 00 00 80 02
0030 ff ff 82 62 00 00 02 04 05 b4 01 03 03 02 01 01
0040 04 02
No.: 53 · Time: 7.075657 · Source: 192.168.1.100 · Destination: 64.233.169.104 · Protoc...4335 → 80 [SYN
Show packet bytes
Layout: Vertical (Stacked)
Help
NAT_ISP_side.pcap
Wireshark · Packet 82 · NAT_ISP_side.pcap
Frame 82: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: Cisco_bf:6c:01 (00:0e:d6:bf:6c:01)
Internet Protocol Version 4, Src: 71.192.34.104, Dst: 64.233.169.104
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0xa2aa (41642)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 127
Protocol: TCP (6)
Header Checksum: 0x04a0 [validation disabled]
[Header checksum status: Unverified]
Source Address: 71.192.34.104
Destination Address: 64.233.169.104
[Stream index: 7]
Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 0, Len: 0
0000 00 0e d6 bf 6c 01 00 08 74 4f 36 23 08 00 45 00
0010 00 34 a2 aa 40 00 7f 06 04 a0 47 c0 22 68 40 e9
0020 a9 68 10 ef 00 50 f8 32 36 e4 00 00 00 00 80 02
0030 ff ff da 46 00 00 02 04 05 b4 01 03 03 02 01 01
0040 04 02
No.: 82 · Time: 6.035475 · Source: 71.192.34.104 · Destination: 64.233.169.104 · Protoc...4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM · SourceIP: 71.192.34.104
Show packet bytes
Layout: Vertical (Stacked)
Help
Close
NAT_home_side.pcap
Packets: 134 · Displayed: 55 (41.0%)
Profile: Default
NAT_ISP_side.pcap
Packets: 210 · Displayed: 55 (26.2%)
Profile: Default
ENG
8:14 PM

4. UDP Stream Analysis in NAT

Applying the filter udp.stream==1 to the home-side capture and udp.stream==0 to the ISP-side capture to isolate the UDP stream.





From above screenshots, home-side and ISP-side

Home-Side

- Source IP: 192.168.1.100
- Destination IP: 68.87.71.230
- Source Port: (DNS ephemeral) same as ISP side
- Destination Port: 53 (DNS)
- Length: same
- Flags: none (UDP has no flags)

ISP-Side

- Source IP: 71.192.34.104
- Destination IP: 68.87.71.230
- Source Port: **same as home-side** (important)
- Destination Port: 53
- Length: same
- The only field NAT changes in UDP is:

Checksum Analysis

Source IP (192.168.1.100 → 71.192.34.104)

Everything in the UDP header (ports, length, payload) stays the same.

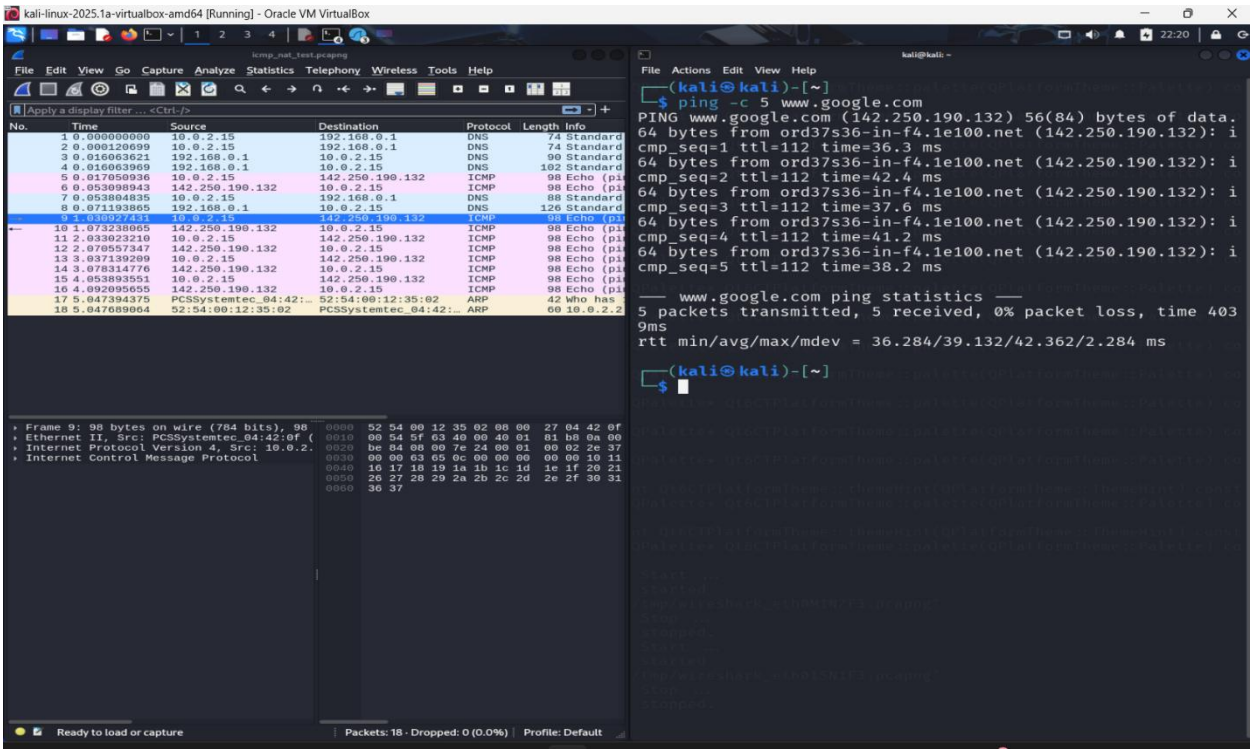
Even though UDP ports and the data are not different, the checksum value of the address + UDP header + data is changed; all that NAT does is modify the IP address.

Because of the UDP pseudo-header, UDP checksum is calculated over below variables

Additional tasks

NAT Timeouts & State Table – TCP Observation

- Wireshark capture interface eth0 on my vm
- I opened TCP connection and let it sit **idle for a long time** so I could see how long the NAT kept the state.



Q1. ICMP Before vs After NAT

ICMP packets used to precede NAT have private source IP (e.g. 10.0.2.15).

Using NAT the source is the public at the ISP side.

The ID field and the Sequence Number field are not altered, as NAT usually only rewrites IP headers unless a checksum needs to be re-calculated.

ICMP checksum is manipulated since the checksum is made of the IP header, and thus rewriting the source IP will require re-calculation.

NAT also alters the source address of the IP, and the ICMP contents remain the same, only the checksum being changed.

Q2. How NAT affects ICMP differently than TCP/UDP

TCP/UDP are port-based means that NAT traces flows utilizing a 5-tuple.

ICMP does not have ports, thus NAT uses the other values which are ICMP Identifier values.

ICMP mappings are easier and have shorter life span since ICMP is usually request only or response only.

NAT can also be more aggressive in dropping ICMP packets as ICMP is not regarded as a connection-oriented protocol.

Q3 How NAT keeps track of active connections & how timeouts impact TCP vs UDP

NAT carries a state table which records the active flows with a 5-tuple of source IP, source port, destination IP, destination port and protocol.

In the case of TCP, NAT entries remain alive on long term since TCP has a distinct hand shake like SYN / SYN-ACK/ACK and tear down FIN/ACK. NAT will update the timestamp as long as there is some traffic in the form of packets.

UDP is different however, it does not have a handshake and as such NAT is not aware of the start and the end of a sessions or of its happenings. UDP mappings have a short expiration duration 60-30 seconds usually.

When the entry does not receive a refresh by any traffic, it is deleted by the NAT, with subsequent UDP packets sent successfully.

Q4 How NAT handles connection re-establishment after state expiry?

In case the NAT entry has expired before the connection completes, any subsequent packet whose state can not be located is regarded as unknown.

NAT can typically , socket books, or drop the packet, which will retransmit the packet to the client making a fresh mapping, which translates as a brand-new connection .

In the case of TCP, this usually induces an entire new three-way handshake.

For UDP, the subsequent packet that leaves the process megarly establishes another connection.

Q5. Filters to Reveal NAT Behavior

ip.src == 10.0.2.15

ip.src == 71.192.34.104

udp.port == 53

tcp.port == 80

kali-linux-2025.1a-virtualbox-amd64 [Running] - Oracle VM VirtualBox

*eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src == 10.0.2.15

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.15	192.168.0.1	DNS	70	Standa
3	0.013507315	10.0.2.15	192.168.0.1	DNS	70	Standa
7	25.755076590	10.0.2.15	142.250.190.78	TCP	74	48906
9	25.797857388	10.0.2.15	142.250.190.78	TCP	54	48906
10	188.362197188	10.0.2.15	142.250.190.78	TCP	54	48906
13	188.400883357	10.0.2.15	142.250.190.78	TCP	54	48906
16	227.428500974	10.0.2.15	1.1.1.1	DNS	47	Unknow
17	419.212366908	10.0.2.15	1.1.1.1	DNS	43	[Malfo
33	4018.3232406…	10.0.2.15	239.255.255.250	UDP/XML	844	56911
34	4018.3296215…	10.0.2.15	224.0.0.22	IGMPv3	54	Member
35	4018.4221889…	10.0.2.15	239.255.255.250	UDP/XML	844	56911
37	4018.6087222…	10.0.2.15	239.255.255.250	UDP/XML	844	56911
38	70694.622448…	10.0.2.15	239.255.255.250	UDP/XML	844	56911
43	70694.913578…	10.0.2.15	224.0.0.22	IGMPv3	54	Member

```
▸ Frame 17: 43 bytes on wire (344 bits), 43
▸ Ethernet II, Src: PCSSystemtec_04:42:0f (
▸ Internet Protocol Version 4, Src: 10.0.2.
▸ User Datagram Protocol, Src Port: 41493,
▾ [Malformed Packet: DNS]
  ▾ [Expert Info (Error/Malformed): Malforn
      [Malformed Packet (Exception occurre
      [Severity level: Error]
      [Group: Malformed]

0000  52 54 00 12 35 02 08 00  27 04 42 0f
0010  00 1d 51 b7 40 00 40 11  db 08 0a 00
0020  01 01 a2 15 00 35 00 09  0e 2b 0a
```

wireshark_eth0B95SF3.pcapn | Packets: 45 · Displayed: 14 (31.1%) · Dropped: 0 (0.0%) | Profile: Default

kali@kali: ~

File Actions Edit View Help

```
┌──(kali㉿kali)-[~]
└─$ ping -c 5 www.google.com
PING www.google.com (142.250.190.132) 56(84) bytes of data.
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=1 ttl=112 time=36.3 ms
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=2 ttl=112 time=42.4 ms
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=3 ttl=112 time=37.6 ms
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=4 ttl=112 time=41.2 ms
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=5 ttl=112 time=38.2 ms

--- www.google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 403
9ms
rtt min/avg/max/mdev = 36.284/39.132/42.362/2.284 ms

┌──(kali㉿kali)-[~]
└─$ nc -lvp 4444
listening on [any] 4444 ...
hello
hi
^C

┌──(kali㉿kali)-[~]
└─$ nc -nv ggogle.com 80
Can't parse ggogle.com as an IP address

┌──(kali㉿kali)-[~]
└─$ nc -nv google.com 80
Can't parse google.com as an IP address

┌──(kali㉿kali)-[~]
└─$ nslookup google.com
Server:         192.168.0.1
Address:        192.168.0.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.191.142
Name:   google.com
```

kali-linux-2025.1a-virtualbox-amd64 [Running] - Oracle VM VirtualBox

*eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

udp.port == 53

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.15	192.168.0.1	DNS	70	Standa
2	0.010794181	192.168.0.1	10.0.2.15	DNS	86	Standa
3	0.013507315	10.0.2.15	192.168.0.1	DNS	70	Standa
4	0.028573817	192.168.0.1	10.0.2.15	DNS	98	Standa
16	227.428500974	10.0.2.15	1.1.1.1	DNS	47	Unknow
17	419.212366908	10.0.2.15	1.1.1.1	DNS	43	[Malfo

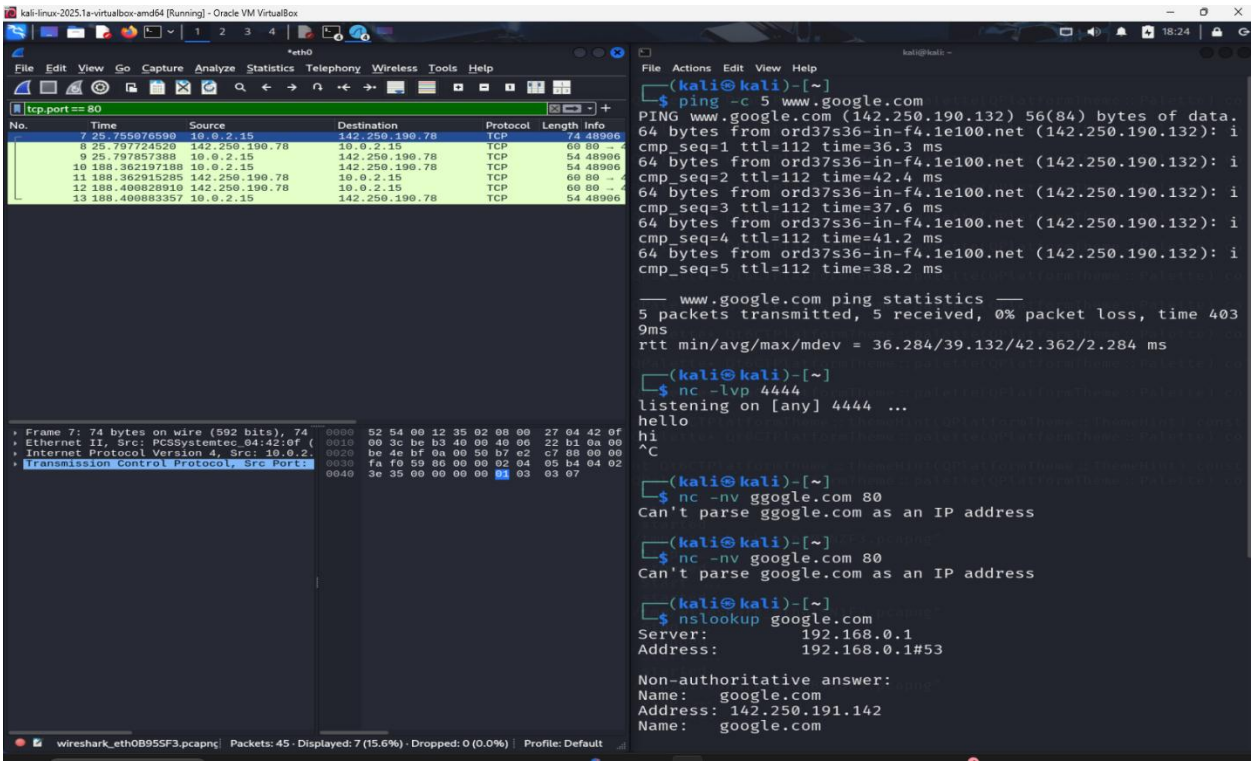
```
▸ Frame 1: 70 bytes on wire (560 bits), 70
▸ Ethernet II, Src: PCSSystemtec_04:42:0f (
▸ Internet Protocol Version 4, Src: 10.0.2.
▸ User Datagram Protocol, Src Port: 36329,
▸ Domain Name System (query)

0000  52 54 00 12 35 02 08 00  27 04 42 0f
0010  00 38 5c f1 00 00 40 11  51 0c 0a 00
0020  00 01 8d e9 00 35 00 24  cc ed 81 ee
0030  00 00 00 00 00 00 06 67  6f 6f 67 6c
0040  6d 00 00 01 00 01
```

wireshark_eth0B95SF3.pcapng | Packets: 45 · Displayed: 6 (13.3%) · Dropped: 0 (0.0%) | Profile: Default

kali@kali: ~

File Actions Edit View Help

```
┌──(kali㉿kali)-[~]
└─$ ping -c 5 www.google.com
PING www.google.com (142.250.190.132) 56(84) bytes of data.
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=1 ttl=112 time=36.3 ms
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=2 ttl=112 time=42.4 ms
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=3 ttl=112 time=37.6 ms
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=4 ttl=112 time=41.2 ms
64 bytes from ord37s36-in-f4.1e100.net (142.250.190.132): i
cmp_seq=5 ttl=112 time=38.2 ms

--- www.google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 403
9ms
rtt min/avg/max/mdev = 36.284/39.132/42.362/2.284 ms

┌──(kali㉿kali)-[~]
└─$ nc -lvp 4444
listening on [any] 4444 ...
hello
hi
^C

┌──(kali㉿kali)-[~]
└─$ nc -nv ggogle.com 80
Can't parse ggogle.com as an IP address

┌──(kali㉿kali)-[~]
└─$ nc -nv google.com 80
Can't parse google.com as an IP address

┌──(kali㉿kali)-[~]
└─$ nslookup google.com
Server:         192.168.0.1
Address:        192.168.0.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.191.142
Name:   google.com
```



Filters here we applied show how packets look before NAT (private IP) and after NAT (public IP), have rewritten addresses and ports.

Q6. How these filters help identify NAT changes

Filtering is used to isolate packets both preceding and succeeding NAT, and fields are easily compared.

It emphasizes what values have been changed:

Source IP

Source port

Checksum

Sometimes TTL

This assists the translation conduct of NAT to be comprehended.

Q7. Compare private IP vs public IP in real-world NAT

The private IP (10.0.2.15) of the VM is the public IP (e.g. 192.168.0.1 - ISP global IP) of the home router.

Here the traces are consistent with the NAT mapping

private IP + private port - Public IP + new public port.

Proving proves that NAT is doing the right thing and restoring IP/port fields leaving destination the same.